HUNT-SAGE · THREAT HUNTING CHEAT SHEET

Network & DNS Hunting (Zeek)

Zeek logs and fields, command-line one-liners, ports of interest and DNS anomaly patterns.

github.com/iamjammie/hunt-sage · iamjammie.github.io/hunt-sage · Built 2026-10-31

Zeek logs that matter

Log	Contents	Hunt uses
`conn.log`	Every connection: 5-tuple, duration, bytes, `conn_state`, `history`	Beaconing, scanning, exfil volume, long sessions
`dns.log`	Queries, answers, rcode, qtype	Tunneling, DGA, rare domains
`http.log`	Method, host, URI, user agent, status, sizes	Rare UAs, downloads, C2 over HTTP
`ssl.log`	SNI, certificate chain, JA3/JA3S (with package), validation	C2 over TLS, self-signed certs
`x509.log`	Certificate details	Cert pivots
`files.log`	File hashes, MIME types, extracted files	Payload delivery
`smb_files.log`, `smb_mapping.log`	SMB file ops, share mounts	Lateral movement, ADMIN$ writes
`dce_rpc.log`	RPC calls	DCSync (`drsuapi`), remote service/task creation (`svcctl`, `atsvc`, `ITaskSchedulerService`)
`kerberos.log`	AS/TGS requests, cipher, success	Kerberoasting, AS-REP roasting
`ntlm.log`	NTLM auth details	PtH, spray, relay
`rdp.log`	RDP cookies, security protocol	RDP from odd places
`ssh.log`	Auth success/fail, client/server strings, direction	Brute force, tunnels
`notice.log`, `weird.log`	Zeek detections, protocol anomalies	Leads

`conn_state` quick guide

State	Meaning
`S0`	SYN seen, no reply (scan, blocked)
`S1`	Established, not closed
`SF`	Normal establish and close
`REJ`	Rejected
`RSTO / RSTR`	Reset by originator / responder
`OTH`	No SYN seen; mid-stream (long-lived / capture started late)

`zeek-cut` one-liners

```
# Top talkers by bytes out
cat conn.log | zeek-cut id.orig_h id.resp_h orig_bytes | awk '{b[$1" "$2]+=$3} END{for(k in b)print b[k],k}' | sort
-rn | head
# Long connections (> 1 hour)
cat conn.log | zeek-cut -d ts id.orig_h id.resp_h id.resp_p duration | awk '$5>3600' | sort -k5 -rn | head
# Rare user agents
cat http.log | zeek-cut user_agent | sort | uniq -c | sort -n | head -30
# DNS: longest queries
cat dns.log | zeek-cut query | awk '{print length($0), $0}' | sort -rn | head
# NXDOMAIN storms per client
cat dns.log | zeek-cut id.orig_h rcode_name | grep NXDOMAIN | cut -f1 | sort | uniq -c | sort -rn | head
# Self-signed / odd certs
cat ssl.log | zeek-cut server_name validation_status | grep -v ok | sort | uniq -c | sort -rn | head
# JA3 stacking (with ja3 package)
cat ssl.log | zeek-cut ja3 | sort | uniq -c | sort -n | head -30
```

Beacon detection tools

- **RITA / AC-Hunter (Active Countermeasures)**: scores beaconing, long connections, DNS tunneling from Zeek logs
- **Custom**: see the timing queries in C2 beaconing

Ports of interest (internal east-west)

Port	Service	Why hunt it
22	SSH	Server-to-server pivoting; tunnels
135, 49152-65535	RPC / DCOM	Remote WMI, scheduled tasks, service creation
139, 445	SMB	PsExec, `ADMIN$`, ransomware spread
3389	RDP	Lateral movement; tunnelled RDP
5985, 5986	WinRM	PowerShell remoting
88	Kerberos	Roasting, ticket abuse
389, 636, 3268, 3269	LDAP / GC	AD recon (BloodHound), LDAP relay
1433, 3306, 5432	Databases	Bulk access, `xp_cmdshell`
53, 853, 443 (DoH)	DNS	Tunneling, resolver bypass
4444, 8080, 8443, 1337, 31337	Common implant defaults	Lead only; C2 uses any port
21, 69, 20	FTP/TFTP	Exfil on legacy protocols

DNS anomaly checklist

- Query length > 50 chars, labels > 30 chars, many unique subdomains of one parent
- High entropy labels (DGA/tunnel); many `TXT` / `NULL` / `CNAME` queries
- High NXDOMAIN ratio per client (DGA)
- Newly registered domains (age < 30 days), rare TLDs (`.top`, `.xyz`, `.click` are overrepresented, but lead only)
- Direct-to-IP DNS to external resolvers; DoH/DoT to non-approved providers
- Queries to dynamic DNS and tunneling services (`ngrok`, `duckdns`, `no-ip`, `trycloudflare`)

PCAP quick triage

```
tcpdump -nn -r cap.pcap 'tcp[tcpflags] & tcp-syn != 0 and tcp[tcpflags] & tcp-ack == 0' | head
tshark -r cap.pcap -q -z conv,tcp | head -30            # conversations by bytes
tshark -r cap.pcap -Y 'http.request' -T fields -e ip.src -e http.host -e http.request.uri -e http.user_agent
tshark -r cap.pcap -Y 'dns.qry.name' -T fields -e ip.src -e dns.qry.name -e dns.qry.type | sort | uniq -c | sort -rn
| head
zeek -r cap.pcap                                        # produce Zeek logs from a pcap
```